ITI SKILLTRACK

Digital Skill Training, Attendance
& Competency Tracking System

PROJECT UNDERSTANDING + SOLUTION BLUEPRINT



FUNCTIONAL DESIGN / SYSTEM ARCHITECTURE / DELIVERY PLAN

Version 1.0 | 30 October 2026

Document control

Item	Details
Document title	ITI SkillTrack - Project Understanding & Solution Blueprint
Working product name	ITI SkillTrack (proposed; final name can be changed)
Document purpose	Translate the supplied problem statement into a complete product, functional, technical and delivery blueprint.
Version / date	Version 1.0 / 30 October 2026
Status	Solution concept and baseline requirements - for stakeholder review
Primary audience	ITI management, instructors/trainers, student representatives, project team and authorized monitoring stakeholders
Source basis	Problem statement and expected outcomes supplied in the request. Implementation details below are proposed design decisions.

IMPORTANT DISTINCTION

The problem statement defines the need and intended outcomes. Screen designs, workflows, data entities, security controls, KPIs, technology options and delivery phases in this document are proposed solution designs, not claims that an existing ITI system already follows them. Validate local institute rules and the applicable trade curriculum before configuration.

Executive summary

ITI SkillTrack is a role-based digital platform for maintaining a connected record of an ITI student's attendance, practical training, competency development, assessment outcomes and progress. It replaces disconnected registers and spreadsheets with a structured workflow and shared source of truth. The system is designed for students, instructors/trainers, institute administrators and authorized monitoring officers.

The core design principle is traceability: every progress indicator should be explainable from the underlying record. Attendance is tied to scheduled sessions; practical completion is tied to defined activities; competency status is tied to criteria and recorded assessment evidence; and remedial support is tied to a specific gap, action, owner and due date.

Current challenge	Proposed capability	Expected operational benefit
Scattered attendance registers and spreadsheets	Central attendance sessions and student-wise records	Fewer duplicate entries; faster retrieval and correction tracking
Practical work difficult to consolidate	Trade-wise activity catalogue and student practical log	Visible activity completion and trainer feedback
Competency status unclear	Competency matrix with criteria, evidence and assessment outcome	More consistent and reviewable skill progress records
Students lack real-time visibility	Student dashboard with attendance, activities and pending requirements	Clearer understanding of progress and next steps
Support needs identified manually	Configurable risk indicators and intervention queue	Earlier, targeted practice and follow-up
Reports assembled by hand	Filterable dashboards and downloadable reports	Faster institutional review and monitoring

What the product must deliver

- One current, authorized record for each student across attendance, training activity and competency assessment.
- A configurable trade, batch and curriculum structure so different ITI trades can use their own modules, activities, criteria and assessment scales.
- Fast instructor workflows for marking attendance, recording practical work, assessing evidence and assigning follow-up.
- A student view that explains what is completed, what is pending, where the student stands and what to do next.
- Institute dashboards and reports that can be filtered by academic session, trade, batch, student and date range.
- Role-based access, audit history, secure evidence storage and operational backup controls.

Document map

Use this map to navigate the complete project definition. The chapter sequence follows the path from problem understanding to design, delivery and validation.

No.	Chapter	Coverage
01	Project overview and context	Purpose, users and operating context
02	Problem analysis and root causes	Current-state pain points and design response
03	Vision, objectives and success measures	Measurable outcomes and KPIs
04	Scope, boundaries and assumptions	MVP, later phases and exclusions
05	Stakeholders, roles and permissions	Role descriptions and access matrix
06	End-to-end workflows	Attendance, practical training, assessment and intervention
07	Functional modules and capabilities	Module-by-module product design
08	Functional requirements catalogue	Testable requirements with priority
09	Competency framework and business rules	Evidence, proficiency, attendance and validation rules
10	Dashboards, analytics and reports	Views, filters, KPI definitions and exports
11	User experience and screen inventory	Information architecture and usability principles
12	Solution architecture and deployment	Logical components and proposed stack
13	Conceptual data model and data dictionary	Core entities, relationships and key fields
14	Security, privacy and governance	Authentication, access, audit and data protection
15	Non-functional requirements and operations	Performance, availability, backup and support
16	Implementation roadmap and deliverables	Phased build plan and project outputs
17	Testing, acceptance and quality assurance	Test coverage and acceptance criteria
18	Risks, dependencies and mitigations	Delivery and operational risk controls
19	Pilot, rollout and change management	Adoption and launch approach
20	Open decisions and glossary	Items to validate and terminology
A	Appendix A - Worked example	Illustrative student journey and competency decision
B	Appendix B - Capture templates	Recommended attendance, practical, assessment and intervention fields
C	Appendix C - Validation rules	Example form validation checks

READING GUIDE

Chapters 1-5 explain the problem and operating model. Chapters 6-15 define how the product works and how it should be built. Chapters 16-20 define how to deliver, test, govern and adopt it.

01 | Project overview and context

The proposed platform is a student training record and institute monitoring system, not merely a digital attendance register.

1.1 Project definition

The Digital Skill Training, Attendance and Competency Tracking System for ITI Students is a web-based application that manages student training information from enrolment through daily attendance, practical activities, assessment and competency review. It presents role-appropriate views while preserving a common, auditable record underneath.

1.2 Operating context

- An ITI typically organizes training around institutes, trades, academic sessions, batches, instructors, theory/practical modules and student cohorts. The system must represent these structures without hard-coding one trade.
- Practical training is activity-oriented. A student may attempt an activity more than once, receive feedback, submit evidence and be assessed against defined criteria.
- Attendance is time/session-oriented. It must be associated with a defined batch session and recorded status, not treated as a free-text percentage.
- Competency is outcome-oriented. Completion of an activity alone should not automatically mean a student is competent unless the configured assessment criteria are met.
- Institute monitoring requires both drill-down (student-level details for authorized staff) and aggregation (trade, batch and period-level summaries).

1.3 Product principles

Principle	Design interpretation
One source of truth	Each key record has one authoritative database record, with controlled edits and audit history.
Evidence before status	Competency outcomes are supported by criteria, assessment and evidence references.
Configurable by trade	Trades, curriculum modules, activities, rubrics and thresholds are master data, not hard-coded logic.
Role-appropriate visibility	Students see their own records; instructors see assigned cohorts; administrators see institute-wide information; monitoring users see approved scope.
Mobile-ready and simple	Frequent tasks use short forms, clear status labels and responsive layouts.
Explainable insights	Alerts and risk indicators show the underlying condition and recommended follow-up; they do not make opaque decisions.

02 | Problem analysis and root causes

The supplied statement identifies operational fragmentation, weak visibility and time-consuming monitoring. These issues are connected and should be solved as one system.

Problem stated	Underlying process gap	Design response
Repeated manual attendance entry	Attendance data is captured in separate registers or spreadsheets; corrections may not be consistently tracked.	Create batch sessions, a digital attendance register, validation rules and an edit audit trail.
Practical records difficult to consolidate	Activities, attempts, observations and evidence lack a common student/trade structure.	Use a trade-wise activity catalogue and student practical log linked to modules and competencies.
No single monitoring dashboard	Data is split across records and reports are prepared separately.	Aggregate trusted records into role-based dashboards with drill-down and export.
Limited student visibility	Students do not have a consolidated personal progress view.	Provide a self-service student portal with attendance, competency, assessment and pending-work views.
Slow identification of students needing support	Risk indicators are not consistently calculated and follow-up may not be assigned.	Create configurable watchlists and intervention records with owner, action, due date and closure.
Manual institutional reporting	Data needs repeated collection, cleaning and consolidation.	Generate consistent reports from the system of record with filters and timestamps.
Need for privacy and controlled access	Shared files or informal workflows can expose records or allow untraceable changes.	Enforce role-based access, secure authentication, audit logs and restricted evidence access.

2.1 What success looks like in daily use

An instructor opens the current batch, marks attendance once, records each student's practical attempt and feedback, and completes an assessment when appropriate. A student can then see the updated record without asking for a manual status check. An administrator can review attendance and competency coverage across trades, identify gaps and export a dated report from the same dataset.

2.2 Why a single attendance-only solution is insufficient

Attendance indicates participation, not demonstrated skill. Practical activity completion indicates exposure or practice, not necessarily competency. Assessment outcomes need criteria and recorded evidence. A credible skill-tracking solution therefore links the chain: scheduled session → attendance → practical activity → evidence → assessment → competency status → follow-up.

03 | Vision, objectives and success measures

3.1 Product vision

> **VISION**
> Give every ITI student and authorized staff member a clear, current and trustworthy view of training participation and demonstrated skill progress - from one secure platform.

3.2 Objectives

1. Digitize daily attendance and reduce repetitive record entry.
2. Maintain a structured trade-wise catalogue of practical activities, competencies and assessment criteria.
3. Give each student self-service access to attendance, completed practicals, assessment history and pending requirements.
4. Enable instructors to assess consistently, provide feedback and assign targeted practice.
5. Give administrators timely institute-level dashboards and downloadable reports.
6. Provide authorized monitoring stakeholders with reliable aggregated information.
7. Protect student data through authentication, authorization, auditability and data-minimization practices.

3.3 Proposed success indicators

MEASUREMENT NOTE

Targets should be agreed after a baseline is measured during discovery or pilot. The values below are proposed pilot indicators, not existing performance figures or guaranteed outcomes.

Indicator	Definition	Suggested pilot target
Attendance capture completeness	Valid attendance records captured / expected student-session records	At least 95% for pilot sessions
Record timeliness	Share of attendance/practical records entered within the institute-approved window	At least 90%
Competency traceability	Assessed competency outcomes with linked criteria and assessor	100% of recorded final outcomes
Student portal adoption	Active student users / students with portal access	At least 80% during pilot
Report preparation effort	Time required to produce an agreed standard report	Reduce effort by 50% from baseline
Intervention closure	Interventions closed by due date / interventions due	Baseline first; improve across review cycles
Data quality	Records passing mandatory-field and relationship checks	At least 98% in pilot data checks

3.4 Outcome chain

Digital capture improves record availability. Linked attendance and training evidence improves visibility. Visibility enables instructors to identify gaps. Assigned remedial practice creates an actionable follow-up loop. Regular review of consistent indicators supports institutional decisions. The system can enable better skill development monitoring, but employment or employability outcomes depend on factors beyond the software and must not be guaranteed by the product.

04 | Scope, boundaries and assumptions

4.1 Minimum viable product (MVP) scope

Area	MVP includes
Institute setup	Institute profile, academic session, trades, batches, student and instructor records, role setup.
Attendance	Session creation, batch register, status marking, validation, correction workflow and attendance summary.
Training activity	Trade/module activity catalogue, activity assignment, student practical log, trainer observation and feedback.
Competency	Competency catalogue, criteria/rubric, assessment record, outcome and student-wise competency status.
Student portal	Personal attendance, activity, assessment, competency and pending-action views.
Dashboards	Student, instructor and administrator dashboards with date/trade/batch filters as authorized.
Reports	Attendance, practical completion, competency, assessment, intervention and consolidated batch reports.
Intervention	Configurable warning/watchlist, remedial action, responsible instructor, due date and closure status.

Area	MVP includes
Security and audit	Authentication, role-based access, audit trail, secure file access and basic backup process.

4.2 Later-phase enhancements

- QR-based session check-in with instructor confirmation and anti-proxy controls appropriate to local policy.
- Offline-first capture for workshops with unreliable connectivity, with conflict-safe synchronization.
- Parent/guardian view, where policy and consent permit.
- Advanced analytics, configurable risk scoring and cohort comparisons over multiple sessions.
- Integration with existing student, examination, attendance or government reporting systems where documented APIs and approvals exist.
- Learning resources, practical video demonstrations, digital portfolio and placement-readiness views.
- Multi-institute or district-level deployment with delegated administration.

4.3 Explicit exclusions unless separately approved

- Biometric attendance, facial recognition, GPS tracking or continuous surveillance.
- Automated pass/fail decisions without a trainer-defined assessment rule and human review.
- Payroll, fees, hostel, inventory, placement or full learning-management functionality.
- Official certification issuance or replacement of any mandated government / awarding-body system.
- Public access to identifiable student information or unrestricted bulk downloads.

4.4 Planning assumptions

Assumption	Validation required
Each institute can provide authoritative student, trade, batch and instructor data.	Confirm source files, data owner, quality and import format.
Trade curricula and practical competencies can be represented as configurable master data.	Confirm current curriculum versions, units, criteria and local assessment terminology.
Attendance categories and calculation rules vary by institute or policy.	Approve status list, session denominator, holidays, leave treatment and correction authority.
Students and staff have access to a browser-enabled device at least periodically.	Validate device mix, language, connectivity and accessibility needs.
Only authorized staff may view institute-wide data.	Approve role assignments and scope boundaries.

05 | Stakeholders, roles and permissions

5.1 User groups

Role	Main responsibility	Typical access boundary
Student	Review own attendance, practical activity, competency, assessment and assigned follow-up.	Own records only; no editing of official outcomes.
Instructor / Trainer	Manage assigned sessions, mark attendance, record practical attempts, assess work and assign practice.	Assigned trade, batch and students.
Trade HOD / Supervisor	Review trade-level progress, oversee instructors, validate exceptions and monitor intervention closure.	Assigned trade(s) and subordinate staff.
Institute Administrator	Manage institute master data, users, academic sessions, batches, reports and approved corrections.	Institute-wide, subject to separation of duties.

Role	Main responsibility	Typical access boundary
Monitoring Officer	Review approved aggregate progress and download authorized reports.	Read-only scope by institute/region/period as granted.
System Administrator	Manage deployment, configuration, technical support and platform security.	Technical administration; access to student content should be minimized and audited.

5.2 Permission matrix (proposed baseline)

Capability	Student	Instructor	HOD	Institute Admin	Monitor
View own student profile / progress	Own	Assigned	Trade scope	Institute	Aggregate or approved scope
Mark attendance	-	Assigned batch	Review / approve exception	Configure / authorize	-
Correct attendance	Request only	Limited, audited window	Approve / correct	Authorized correction	-
Record practical activity	-	Assigned students	Review	Configure master data	-
Assess competency	-	Assigned competencies	Moderate / review	Configure rubrics	-
Assign remedial plan	View / acknowledge	Create / update	Oversee	Review	Aggregate only
Manage users and batches	-	-	Limited delegation	Yes	-
View institute dashboard	-	Assigned batch	Trade scope	Yes	Approved scope
Export student-level report	Own summary only	Assigned scope	Trade scope	Yes	Only if explicitly authorized
View audit log	-	Own actions where needed	Trade actions	Institute actions	No or aggregate

AUTHORIZATION PRINCIPLE

Apply least privilege and deny-by-default. The permission matrix is a starting point; the final matrix must be approved by institute leadership and translated into server-side authorization checks for every API and export. Hiding a button in the interface is not access control.

5.3 Separation of duties

A user who enters a record should not automatically be able to approve every exception to that same record. For example, attendance correction after a defined lock window may require HOD or administrator approval. Final competency outcomes may require the assigned assessor, with moderation rights delegated to a supervisor. The system should record both the original value and the approved correction.

06 | End-to-end workflows

The platform links operational records into a repeatable training lifecycle. Each step has an owner, a record and a validation point.

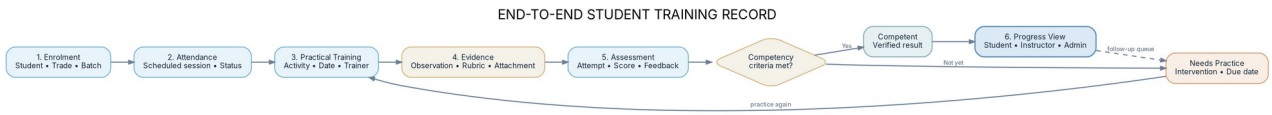


Figure 1. Proposed workflow from enrolment through practice, assessment and intervention.

6.1 Master setup and enrolment

1. Administrator creates academic session, trade, batch and curriculum version, then maps instructors to batches/modules.
2. Student records are created or imported, checked for duplicate identifiers and assigned to the appropriate batch.
3. Enrollment status (active, transferred, completed, withdrawn, inactive) controls whether the student appears in current registers.

6.2 Attendance workflow

1. Instructor selects batch, date and scheduled period/session. The system loads the expected active student list.
2. Instructor marks each student using the approved status list. Bulk actions may be provided, but individual exceptions must remain visible.
3. The system validates duplicates, missing statuses, session date and instructor authorization, then saves with actor and timestamp.
4. Changes after a configurable lock period require a reason and approval. The record retains a correction history.
5. Dashboards recompute attendance using the institute-approved denominator and exclusion rules.

6.3 Practical training and competency workflow

1. Instructor selects a curriculum module and practical activity relevant to the trade and cohort.
2. Student attempt is recorded with date, activity, trainer, status, observations, feedback and optional evidence attachment.
3. When the student is ready, an assessor evaluates one or more linked competencies against configured criteria or a rubric.
4. The assessor records criterion-level result, score where applicable, outcome, feedback and evidence reference.
5. The system updates competency status only according to the configured rule; an activity marked complete is not automatically a competent result.
6. If criteria are not met, the instructor creates a remedial action with practice task, owner, due date and review outcome.

6.4 Intervention workflow

A student may enter a watchlist because of low attendance, an overdue practical, repeated unsuccessful attempts or an instructor-raised concern. The platform should show why the student is flagged, allow staff to verify the underlying data, and create a follow-up action. The flag is a support prompt, not a disciplinary decision or a substitute for professional judgement.

6.5 Typical record lifecycle

Record	Lifecycle states (configurable)	Control point
Attendance session	Draft → Open → Submitted → Locked → Amended (audited)	Session date, batch, completeness and approval for late correction.
Practical log	Assigned → In Progress → Submitted → Reviewed → Completed / Rework	Trainer confirmation and activity-specific completion conditions.
Assessment	Scheduled → Attempted → Evaluated → Moderated (optional) → Final	Assessor identity, criteria, outcome and evidence.
Competency	Not Started → In Progress → Needs Practice → Competent	Competent requires all mandatory criteria or approved rule.
Intervention	Open → In Progress → Awaiting Review → Closed / Escalated	Owner, due date, review note and closure reason.

07 | Functional modules and capabilities

Module	Core capabilities	Primary users
M01 - Identity, authentication and roles	Secure sign-in, password reset, account activation/deactivation, role assignment, session management and access checks. Support account lockout/rate limiting and administrator-controlled reset.	All users; system admin
M02 - Institute and master data	Institute profile, academic sessions, trades, batches, divisions, curriculum versions, modules, competencies, practical activities, assessment rubrics, holidays and instructor mappings.	Institute admin; HOD
M03 - Student and enrolment management	Student profile, identifiers, batch/trade enrolment, status changes, imports, duplicate checks and controlled profile updates.	Institute admin; limited instructor view
M04 - Attendance management	Session timetable or session creation, digital register, bulk mark, status validation, absent list, correction requests, approval and attendance summaries.	Instructor; HOD; admin; student view
M05 - Practical training log	Activity library, assigned work, student attempt, trainer observation, feedback, completion status, evidence attachment and repeat attempt history.	Instructor; student read view
M06 - Competency framework	Competency catalogue, criteria, rubric, required evidence, proficiency scale, assessment mapping and competency status history.	HOD/admin configure; instructor assess
M07 - Assessment and evaluation	Assessment event, assessor, attempt, criterion results, score/grade where applicable, outcome, comments, moderation and result publication.	Instructor/assessor; HOD review
M08 - Student progress portal	Personal dashboard, attendance trend, activity completion, competency matrix, assessment history, pending tasks and intervention status.	Student
M09 - Dashboards and analytics	Role-based cards, trend charts, filters, watchlists, cohort comparison and drill-down. All metrics trace back to source records.	Instructor; HOD; admin; monitor
M10 - Intervention and remedial support	Configurable triggers, staff review, assigned practice plan, owner, due date, follow-up note, outcome and escalation.	Instructor; HOD; student view
M11 - Reports and exports	Standard PDF/CSV/Excel-compatible exports, report parameters, generation timestamp, data scope and export audit.	Authorized staff
M12 - Notifications and reminders	In-app alerts for missing attendance, pending activity, assessment result, intervention due date and report-ready events. External SMS/email optional.	All roles by preference and policy
M13 - Audit and system settings	Audit search, configuration history, backup status, data import logs, retention settings and access review support.	Institute/system administrator

7.1 Module dependencies

M02 and M03 establish the master records needed by all operational modules. M04 depends on an active batch and session. M05 depends on curriculum/activity setup. M06 and M07 depend on competencies and assessment criteria. M08, M09 and M11 read from the operational records; M10 uses their outcomes to coordinate support. This dependency order should drive implementation sequencing.

08 | Functional requirements catalogue

The following requirements are testable baseline requirements. MoSCoW priority is proposed: Must = required for the MVP; Should = important but can follow the first release; Could = optional enhancement.

ID	Requirement	Priority
FR-01	System shall authenticate each user with an individual account.	Must
FR-02	System shall enforce role- and scope-based authorization server-side for screens, APIs, records and exports.	Must
FR-03	Administrator shall configure academic sessions, trades, batches and curriculum versions.	Must
FR-04	Administrator shall create, import, update and deactivate student and instructor records with duplicate validation.	Must
FR-05	Instructor shall create or open an attendance session for an authorized batch and date.	Must
FR-06	System shall display the expected active student roster for the selected session.	Must
FR-07	Instructor shall mark approved attendance statuses and submit the register.	Must
FR-08	System shall prevent duplicate student attendance records for the same session.	Must
FR-09	System shall record the user, timestamp and reason for attendance corrections.	Must
FR-10	Authorized approver shall approve or reject corrections after the lock window.	Should
FR-11	Administrator shall maintain trade-wise modules, practical activities and competency criteria.	Must
FR-12	Instructor shall record practical attempts, observations, feedback and status for assigned students.	Must
FR-13	System shall support multiple attempts for a practical activity without overwriting prior attempts.	Must
FR-14	Assessor shall evaluate a competency against configured criteria/rubric.	Must
FR-15	System shall retain assessor, date, outcome, comments and evidence references for each assessment.	Must
FR-16	System shall calculate competency status only from configured completion/assessment rules.	Must
FR-17	Student shall view their own attendance, practical log, assessment and competency status.	Must
FR-18	Student shall see pending requirements and assigned remedial actions.	Must
FR-19	Instructor shall view assigned batch dashboard and student drill-down.	Must
FR-20	HOD/admin shall view authorized trade/institute dashboards with filters.	Must

ID	Requirement	Priority
FR-21	System shall calculate attendance percentage using an approved, documented denominator.	Must
FR-22	System shall create watchlist entries using configurable rules and allow staff verification.	Should
FR-23	Staff shall assign a remedial action with owner, action, due date and closure note.	Must
FR-24	System shall generate standard attendance, competency, assessment and intervention reports.	Must
FR-25	Authorized users shall export reports in approved formats with scope and timestamp.	Must
FR-26	System shall maintain searchable audit events for sensitive create/update/delete/export actions.	Must
FR-27	System shall store uploaded evidence outside public web paths and authorize each download.	Must
FR-28	System shall support data import validation and provide row-level error feedback.	Should
FR-29	System shall provide in-app notifications for key pending actions.	Should
FR-30	System shall support optional email/SMS notifications through a configurable adapter.	Could
FR-31	System shall provide an accessible, responsive interface for common mobile and desktop sizes.	Must
FR-32	System shall support data export and documented backups to reduce operational lock-in.	Must

REQUIREMENT GOVERNANCE

Before development, assign an owner and acceptance test to every Must requirement. Changes to roles, attendance calculations, assessment scales and report definitions should be versioned and approved because they affect institutional records.

09 | Competency framework and business rules

9.1 Curriculum configuration model

The system should model a hierarchy that can be adapted to the institute's actual curriculum structure: Trade → Curriculum Version → Module / Unit → Competency → Assessment Criteria → Practical Activity / Evidence. The exact labels and levels must be configurable and aligned with the curriculum version adopted by the institute.

Level	Example fields	Why it exists
Trade	Trade code, name, active flag	Separates trade-specific training structures.
Curriculum version	Version label, effective dates, reference document	Preserves which curriculum was applicable to a batch.
Module / unit	Code, title, sequence, hours (optional)	Groups learning and practical activities.
Competency	Code, statement, expected level, mandatory flag	States the skill outcome to be demonstrated.

Level	Example fields	Why it exists
Assessment criteria	Criterion text, evidence type, scoring method, pass rule	Makes evaluation explainable and repeatable.
Practical activity	Title, instruction, safety note, equipment (optional), mapped competencies	Defines work a student can perform and log.
Evidence	Observation, checklist, file, result, date, assessor	Supports an auditable competency decision.

9.2 Suggested competency status model

Status	Definition	Permitted transition
Not Started	No recorded attempt or evidence for this competency.	→ In Progress
In Progress	Student has started related learning or practice; evidence is incomplete.	→ Needs Practice / Ready for Assessment
Ready for Assessment	Required practice/evidence has been submitted for review.	→ Competent / Needs Practice
Needs Practice	Assessment did not meet one or more required criteria, or trainer requested further practice.	→ In Progress / Ready for Assessment
Competent	All mandatory criteria are met under the configured rule and an authorized assessor records the outcome.	Reassessment or superseding result only through controlled process

CONFIGURABLE SCALE

The suggested status names are a product-level workflow. They must not be confused with any official grade, certification or regulatory proficiency label. Where the institute uses a different assessment scale, configure that scale and preserve the mapping/version.

9.3 Assessment design

- Each assessment should identify the competency, criteria, assessor, date, attempt number, outcome and feedback.
- Criteria may be binary (met/not met), rubric-based (for example, levels defined by the institute) or numeric. The scale should be selected per assessment type, not forced globally.
- Mandatory criteria and pass rules should be explicit. A weighted score must not override a failed mandatory safety or quality criterion unless the approved rule says so.
- Reassessment creates a new attempt. Prior outcomes remain available for audit and longitudinal progress.
- Attachments should be optional and purpose-limited. In many practical assessments, a trainer checklist and observation may be sufficient.

9.4 Core business rules

Rule ID	Rule
BR-01	A student must have one active enrollment mapping to the applicable trade/batch for a given period; transfers are recorded as dated history.
BR-02	An attendance record belongs to exactly one scheduled session and one student; duplicate student-session pairs are rejected.
BR-03	Attendance percentage = count of attended eligible sessions / count of eligible scheduled sessions x 100, subject to the approved status mapping and exclusions.

Rule ID	Rule
BR-04	Cancelled sessions and approved exclusions must be handled consistently and must not silently inflate or reduce attendance.
BR-05	Only authorized assigned staff may mark attendance or assess a student. Corrections require a reason and create an audit event.
BR-06	Practical activity completion and competency attainment are separate fields and states.
BR-07	A competency becomes Competent only when all configured mandatory criteria are met and the assessment is finalized by an authorized assessor.
BR-08	An intervention must have a reason, assigned owner, action and review date; closing it requires an outcome note.
BR-09	Reports must state the applied date range, filters, scope, generation timestamp and calculation definitions.
BR-10	A curriculum or rubric change must be versioned; historical student records retain the version used when assessed.

10 | Dashboards, analytics and reports

10.1 Role-based dashboards

Dashboard	Recommended content	Primary actions
Student home	Attendance percentage and trend; practical activities complete/in progress; competency matrix; recent assessment results; pending practice; due dates.	Open an activity; read feedback; acknowledge a remedial task; view report card.
Instructor home	Today's sessions; attendance pending; assigned batches; practical logs awaiting review; assessments due; students on watchlist.	Open register; review evidence; assess; add feedback; assign intervention.
HOD / Supervisor	Trade-wise attendance; competency distribution; activity backlog; assessment completion; intervention ageing; instructor coverage.	Filter by batch/module; review exceptions; monitor action closure.
Institute administrator	Institute summary; active students/batches; attendance and competency trend; data completeness; import status; report centre.	Manage master data; authorize corrections; export reports; review audit exceptions.
Monitoring officer	Approved aggregated indicators by institute/trade/period; comparative trends where permitted.	Filter and export approved summaries; no default student-level access.

10.2 KPI definitions

KPI	Recommended calculation / definition	Caution
Attendance %	Eligible attended sessions / eligible scheduled sessions x 100	Publish status mapping and treatment of leave, holidays and cancelled sessions.
Practical completion %	Activities marked complete / activities assigned in selected scope x 100	Define whether activities are mandatory, optional or weighted.
Competency coverage %	Competencies with finalized status / competencies expected for the selected scope x 100	Do not count Not Started as completed; show Needs Practice separately.

KPI	Recommended calculation / definition	Caution
Competent rate	Competencies in Competent status / competencies assessed or expected, as explicitly selected	Show denominator and avoid mixing not-yet-assessed with unsuccessful results.
Assessment completion	Assessments finalized / assessments scheduled or assigned	Define treatment of cancelled and rescheduled assessments.
Intervention overdue	Open interventions past due date	A watchlist is for support, not a punitive ranking.
Data completeness	Records with all required fields / records in the selected dataset	Use validation rules, not manual assumptions.

10.3 Standard reports

- Student progress report: profile, trade/batch, attendance by period, practical activities, competency outcomes, assessment history and pending actions.
- Daily attendance register: batch roster, session/date, status, marked by, submission time and corrections.
- Monthly attendance summary: student-wise percentage, eligible session count and configurable status breakdown.
- Trade/batch competency matrix: competencies as rows or columns, students as rows or columns, with status and last assessment date.
- Practical activity completion report: assigned, attempted, completed, rework and overdue counts.
- Assessment result register: assessment, competency, assessor, attempt, criteria outcome, score if applicable and feedback.
- Remedial intervention report: reason, student, owner, created date, due date, status, ageing and closure outcome.
- Institute monitoring summary: aggregated participation, competency coverage, pending assessments and data completeness.

> **REPORTING CONTROL**
> Every export should inherit the user's access scope, include filter context and generation time, and be logged. Avoid exposing unrelated student personal details in summary reports.

11 | User experience and screen inventory

11.1 Navigation structure

Role	Primary navigation
Student	Home | My Attendance | My Practical Work | My Competencies | Assessments | My Action Plan | Profile
Instructor	Dashboard | My Batches | Attendance Register | Practical Activities | Assessments | Student Progress | Interventions
HOD	Trade Dashboard | Batches | Competency Matrix | Attendance Review | Intervention Queue | Reports
Institute Admin	Institute Dashboard | Students | Staff | Trades & Curriculum | Sessions & Batches | Users & Roles | Reports | Audit & Settings
Monitoring	Overview | Institute / Trade Summary | Trends | Approved Exports

11.2 Key screens

Screen	Essential information and interaction
Login / account recovery	Institution branding, sign-in, password recovery, clear account support message.
Student dashboard	Progress cards, recent updates, attendance trend, activity status, competency overview and next action.
Attendance register	Batch/date/session selector, roster, status controls, bulk mark, incomplete-state warning, save/submit and lock state.
Practical activity detail	Activity instructions, competency mapping, student attempt list, evidence, observations, feedback and review action.
Assessment workspace	Criteria list/rubric, evidence preview, attempt number, outcome, comments, save draft and finalize confirmation.
Competency matrix	Trade/module filters, student rows, competency columns/statuses, last assessed date and drill-down.
Intervention workspace	Reason, source metric, student context, assigned action, owner, due date, follow-up and closure.
Report centre	Report type, period, trade/batch, student scope, preview, download and report definition.
Master data setup	Versioned trade, batch, curriculum, activity and rubric configuration with effective dates and validation.

11.3 UX standards

- Prioritize common instructor tasks: select batch, mark register, save and move to next task with minimal navigation.
- Use clear labels and consistent status colors, but never rely on color alone; include text and accessible contrast.
- Provide search, sort, pagination and bulk actions for long student lists. Confirm destructive or finalizing actions.
- Keep forms short and progressively reveal optional fields such as evidence attachments.
- Display data freshness, date range and calculation method beside dashboard indicators.
- Support keyboard navigation, meaningful focus order, readable typography and responsive layouts.
- Use plain language in student views; expose staff-level technical detail only where it helps a task.

12 | Solution architecture and deployment

The architecture below is a reference design. It can be implemented with the institute-approved technology stack; the core requirement is separation of presentation, business rules, persistence and file access.

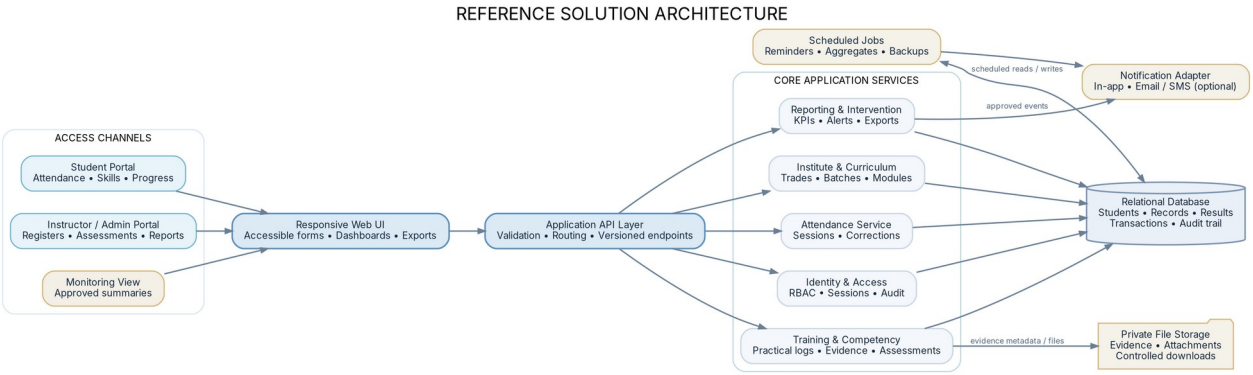


Figure 2. Logical architecture for a responsive web application with controlled API and data layers.

12.1 Architectural layers

Layer	Responsibilities	Design constraints
Presentation layer	Responsive web UI for students, instructors, administrators and monitoring users.	No business-critical permission or calculation should live only in the browser.
API / application layer	Request validation, authentication, authorization, workflow orchestration and consistent error responses.	Use server-side validation and versioned endpoints; avoid direct database access from UI.
Domain services	Attendance, enrolment, curriculum, practical activity, assessment, competency, intervention and reporting rules.	Keep rules centralized and testable; use transactions for linked updates.
Persistence layer	Relational database, migrations, indexes, constraints and query layer.	Enforce referential integrity and uniqueness at database level where practical.
File service	Private evidence upload, metadata, access authorization, size/type validation and retention.	Store outside public web root or in private object storage; serve via authorized handler.
Integration adapters	Email/SMS, imports, optional QR or other approved external systems.	Adapters are optional; do not tightly couple core workflows to external providers.
Operations layer	Logging, monitoring, scheduled tasks, backups, restore testing and deployment.	Protect secrets; separate dev/test/production and document recovery steps.

12.2 Suggested implementation profile

TECHNOLOGY PROPOSAL - NOT A FIXED REQUIREMENT

For a straightforward institute deployment, a practical baseline is HTML5, CSS3 and JavaScript for the responsive interface; PHP 8+ for the server/API; MySQL 8+ for relational storage; and Apache or Nginx over HTTPS. A frontend framework can be added if the delivery team already uses one. Final versions, hosting, licensing and security posture must be reviewed before procurement/deployment.

12.3 Deployment topology

- Production traffic reaches the application through HTTPS and a reverse proxy/web server.
- The application runtime accesses the database using a least-privileged service account; the database is not exposed directly to the public internet.
- Uploaded evidence is stored in private storage with non-guessable identifiers and controlled download routes.
- Scheduled jobs handle reminders, summary aggregation, data housekeeping and backup orchestration; they must be observable and retry-safe.
- Development, testing/staging and production environments use separate credentials and datasets. Production student data should not be copied into development without approved masking.

12.4 API design examples

Method / endpoint (illustrative)	Purpose	Authorization
GET /api/v1/batches/{id}/attendance?date=YYYY-MM-DD	Retrieve register for a session/date.	Instructor assigned to batch or authorized admin.
POST /api/v1/attendance-sessions	Create a batch attendance session.	Authorized instructor/admin.
PUT /api/v1/attendance-records/{id}	Update a record within permitted correction window.	Assigned instructor; approver for exception flow.
POST /api/v1/practical-logs	Create a practical attempt/observation.	Assigned instructor or permitted student submission.

Method / endpoint (illustrative)	Purpose	Authorization
POST /api/v1/assessments/{id}/finalize	Finalize criterion-level assessment.	Authorized assessor; optional moderation rule.
GET /api/v1/students/me/progress	Return authenticated student progress.	Student account; server derives student identity.
GET /api/v1/reports/attendance	Generate or retrieve a scoped report.	Role and scope checked before query/export.

API SECURITY RULE

Never trust a student_id, institute_id, trade_id or role sent by the browser as proof of access. Derive identity and permitted scope from the authenticated account and validate every requested record server-side.

13 | Conceptual data model and data dictionary

The model is relational and history-aware. Table and field names are illustrative; the final schema should be normalized and reviewed against actual reporting and curriculum needs.

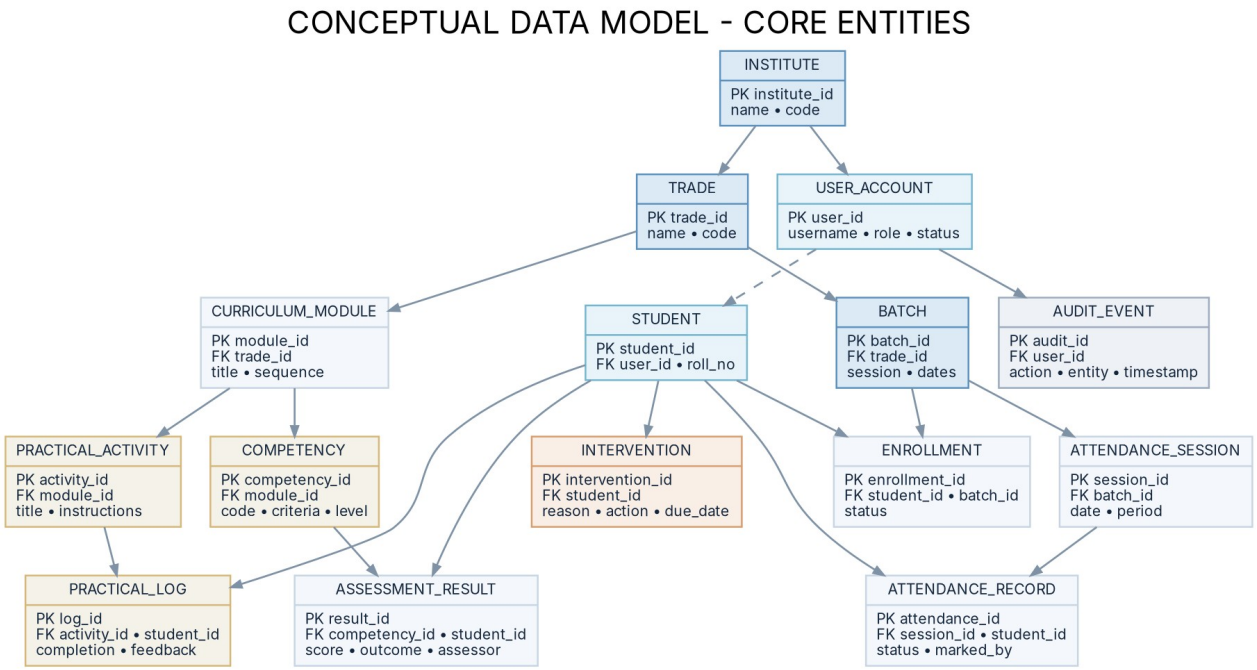


Figure 3. Conceptual entity relationships. The production schema will add timestamps, constraints, status history and audit metadata.

13.1 Core entities

Entity	Purpose	Important relationships
Institute	Tenant / institute boundary and institutional profile.	Owns trades, users, sessions, batches and settings.
UserAccount	Login identity, status, authentication and role mapping.	May map to student or staff profile; creates audit events.
Student	Student identity and institute-specific profile.	Linked to enrollment, attendance, practical logs, assessments and interventions.
Trade	Trade master and code.	Owns curriculum versions/modules and batches.
Batch	Student cohort, academic period and trade mapping.	Contains enrolments; receives attendance sessions.

Entity	Purpose	Important relationships
Enrollment	Dated relationship between student and batch.	Supports transfer and historical cohort reporting.
CurriculumModule	Trade/module/unit under a curriculum version.	Contains competencies and practical activities.
Competency	Expected skill outcome and its criteria.	Assessed through assessment results; linked to activities.
PracticalActivity	Defined practical work or task.	Generates multiple student practical log attempts.
AttendanceSession / Record	Scheduled attendance event and each student status.	Unique student-session record; linked to batch and marker.
PracticalLog	Student attempt, observation, feedback and completion.	Links student, activity, trainer and optional evidence.
AssessmentResult	Criterion/assessment outcome for a competency attempt.	Links student, competency, assessor, rubric version and evidence.
Intervention	Remedial/follow-up action and closure status.	Links student, reason/source, owner and review outcome.
AuditEvent	Append-only record of sensitive actions.	Links actor, entity type/id, action, timestamp and change reference.

13.2 Suggested field groups

Record	Representative fields
Student	student_id, institute_id, admission_no, full_name, contact fields (minimum required), status, created_at, updated_at
Batch	batch_id, trade_id, academic_session_id, batch_code, start_date, end_date, status
Enrollment	enrollment_id, student_id, batch_id, effective_from, effective_to, enrollment_status, change_reason
AttendanceSession	session_id, batch_id, session_date, period_code, start_time, end_time, session_status, created_by, submitted_at, locked_at
AttendanceRecord	attendance_id, session_id, student_id, attendance_status, remarks, marked_by, marked_at, corrected_by, correction_reason
PracticalLog	log_id, student_id, activity_id, attempt_no, trainer_id, activity_date, status, observations, feedback, evidence_id
AssessmentResult	result_id, student_id, competency_id, rubric_version_id, attempt_no, assessor_id, assessed_at, outcome, score, feedback, finalized_at
Intervention	intervention_id, student_id, trigger_type, source_record_id, action_plan, assigned_to, due_date, status, closure_note
AuditEvent	audit_id, actor_user_id, action, entity_type, entity_id, occurred_at, source_ip (where appropriate), correlation_id

13.3 Data integrity and history

- Use stable primary keys and foreign keys. Add unique constraints for student admission identifiers within their defined institute scope and for student-session attendance pairs.
- Use effective dates or version records for trade curricula, rubrics and student batch enrollment so historical reports remain interpretable.
- Keep timestamps in a consistent server-side convention and display them in the institute local timezone.
- Use soft-deactivation for master records that are referenced by historical records; do not cascade-delete student training history.

- Keep evidence metadata in the database and file bytes in private storage. Apply file size, MIME/type and malware-scanning controls appropriate to the environment.
- Use transactions for multi-step actions such as finalizing an assessment and updating its derived competency state.

14 | Security, privacy and governance

14.1 Security baseline

Control area	Minimum design requirement
Authentication	Individual accounts; strong password policy; secure password hashing using a maintained algorithm; rate limiting; controlled recovery; optional MFA for privileged users.
Authorization	RBAC plus data scope (institute/trade/batch/assigned student); deny by default; server-side checks on every endpoint and export.
Transport and storage	HTTPS; protected secrets; encrypted backups; private evidence storage; avoid placing credentials or personal data in logs.
Input and application security	Parameterized SQL; server-side validation; output encoding; CSRF protection for cookie sessions; secure headers; safe upload validation.
Session security	Secure, HttpOnly and appropriate SameSite cookies; session rotation after login/privilege change; expiry and logout invalidation.
Auditability	Record sensitive actions including create/update/correction/finalization/export, actor, timestamp, target record and reason where relevant.
Monitoring	Centralized error/security logs, alerting for unusual login and export activity, and restricted access to logs.
Backup and recovery	Encrypted backups, documented retention, restricted backup access and periodic restore testing.

14.2 Privacy-by-design requirements

- Collect only student personal information required for enrolment, communication, training administration and approved reporting.
- Show student-level data only to roles with a defined educational or administrative need.
- Keep monitoring dashboards aggregated by default; grant identifiable drill-down only when explicitly authorized.
- Define retention and deletion/archival rules with institute policy and applicable law before production use.
- Explain to users what information is recorded, who can view it and how corrections are requested.
- Do not introduce biometrics, facial recognition, GPS or sensitive profiling as a default feature. Any such capability requires separate necessity, legality, proportionality and approval review.
- Provide a correction workflow for inaccurate student records and preserve the approval/audit history.

14.3 Governance and operational ownership

Governance item	Proposed owner	Required decision / artefact
Student master data	Institute administration	Authoritative fields, unique identifier and import approval.
Curriculum and rubric	Trade HOD / academic owner	Approved curriculum version, criteria and effective dates.
Attendance policy	Institute leadership / designated authority	Status mapping, denominator, lock window and correction approval.

Governance item	Proposed owner	Required decision / artefact
Role and access review	Institute administrator	User-role matrix and periodic access recertification.
Data retention	Institute leadership / data owner	Retention schedule, archival and disposal process.
Security operations	Technical administrator / hosting provider	Patch process, backup, incident response and recovery responsibilities.
Report definitions	Academic / monitoring owner	Approved formulas, filters, aggregation and report distribution rules.

GOVERNANCE CAUTION

The institute should obtain legal and policy review for applicable student-data, cybersecurity, retention and government-hosting requirements before go-live. This blueprint is a technical design aid, not legal advice or a certification of compliance.

15 | Non-functional requirements and operations

Quality attribute	Proposed baseline requirement	Verification method
Usability	An instructor can mark a normal batch register with a small number of interactions and without repeated re-entry.	Observed task test with representative instructors.
Responsive layout	Core tasks work on current desktop and mobile browser sizes; no horizontal scrolling for primary forms.	Viewport and device testing.
Performance	Common list/dashboard views should return within an agreed target under pilot load; target 95th percentile under 3 seconds for typical queries.	Load test with realistic data volume.
Availability	Agree business-hour availability and maintenance windows; proposed MVP target 99.5% monthly excluding planned maintenance, subject to hosting.	Monitoring and monthly availability report.
Scalability	Support growth in trades, batches, students, sessions and evidence files without redesigning core entities.	Volume test and index/query review.
Reliability	Database constraints and transactions prevent partial or duplicate official records.	Failure injection and transaction tests.
Security	Pass an application security review before production; resolve critical/high findings before launch.	Threat model, static/dynamic testing and code review.
Accessibility	Keyboard operable, clear labels, sufficient contrast and semantic structure for common tasks.	Accessibility audit and assistive-technology spot checks.
Maintainability	Documented API, schema migrations, configuration, tests and deployment procedure.	Engineering review and deployment rehearsal.
Interoperability	CSV import/export with documented templates; API integration only where approved documentation exists.	Import/export and contract tests.
Auditability	Sensitive records can be traced to an authorized actor and timestamp.	Audit trail sampling.
Recoverability	Backups run on an agreed schedule and a restore procedure is tested.	Scheduled restore drill.

15.1 Operational checklist

- Assign an operational owner for application, database, file storage, domains, certificates and backup monitoring.
- Patch the operating system, runtime, framework and dependencies on a documented cadence.
- Keep separate environments and prohibit production credentials in source code or repositories.
- Monitor disk capacity, database growth, failed jobs, error rates, authentication failures and backup completion.
- Review user access at batch/session changes and at least at an institute-approved periodic interval.
- Run restoration drills and document recovery time and recovery point objectives before production reliance.
- Provide a support path for incorrect records, locked accounts, report problems and device/connectivity issues.

16 | Implementation roadmap and deliverables

The sequence below is a planning baseline for an MVP. Actual duration depends on team size, data quality, hosting approvals, curriculum complexity and integration requirements.

Phase	Indicative duration	Key work	Exit deliverables
0. Discovery and validation	1-2 weeks	Confirm users, policies, attendance formula, curriculum/rubric structure, source data, reports and hosting constraints.	Approved requirement baseline; process maps; data inventory; decision log.
1. Product and UX design	1-2 weeks	Information architecture, role matrix, screen wireframes, design system, API/data model and security threat review.	Approved prototype; schema; architecture; acceptance criteria.
2. Platform foundation	2 weeks	Environment, authentication, roles, institute setup, master data, import validation and audit framework.	Working login, RBAC, admin setup, schema migrations, import template.
3. Attendance and student records	2 weeks	Enrolment, batch roster, session creation, marking, corrections and attendance reporting.	Attendance MVP and test evidence.
4. Practical and competency modules	2-3 weeks	Activity catalogue, practical log, assessment rubric, competency status and intervention workflow.	Training/competency MVP; assessor and remediation workflow.
5. Dashboards, reports and hardening	1-2 weeks	Student/instructor/admin dashboards, exports, performance, security and accessibility fixes.	Release candidate; report pack; test results.
6. Pilot and rollout	2-4 weeks	Data load, user training, pilot, issue resolution, sign-off and phased launch.	Pilot report; training materials; production deployment; support plan.

16.1 Delivery workstreams

Workstream	Tasks / artefacts
Product & process	Requirements, workflows, policy decisions, acceptance criteria, change control.
UX & content	Role navigation, responsive layouts, status language, accessibility, help text and training material.
Application engineering	Frontend, API, services, data schema, imports, report generation and automated tests.
Data preparation	Student, trade, batch and curriculum data cleanup, mapping, duplicate resolution and migration reconciliation.
Security & operations	Threat model, secrets, environments, backups, monitoring, restore test and operational runbook.

Workstream	Tasks / artefacts
Pilot & adoption	Representative pilot cohort, instructor training, student onboarding, feedback and launch readiness.

16.2 Suggested project deliverables

- Project charter, validated scope, stakeholder register and decision log.
- Functional requirements specification and acceptance criteria.
- UI wireframes/prototype and screen inventory.
- System architecture, database schema, data dictionary and API specification.
- Security and privacy assessment, backup/recovery runbook and access matrix.
- Working MVP with source code, migrations, configuration template and deployment instructions.
- Test plan, test cases, test execution evidence, defect log and UAT sign-off.
- User guides for students, instructors, administrators and monitoring users.
- Data migration report, pilot review, go-live checklist and support ownership plan.

17 | Testing, acceptance and quality assurance

17.1 Test strategy

Test level	Coverage	Examples
Unit tests	Individual business rules and validation functions.	Attendance percentage; competency completion rule; role-scope resolver; duplicate session record check.
Integration tests	API, database, file storage and scheduled job interactions.	Save attendance transaction; upload/download authorization; finalize assessment and update competency.
System tests	End-to-end user journeys across roles.	Instructor marks attendance; student sees update; admin exports batch summary.
Security tests	Authentication, authorization, injection, CSRF, file handling, session and export scope.	Student attempts to access another student ID; instructor requests another batch; unauthorized evidence download.
Performance tests	Realistic cohort size, register operations, dashboard aggregation and report generation.	Concurrent instructors marking registers; large trade competency matrix.
Usability/accessibility tests	Common task efficiency, clarity, keyboard use, responsive behavior and contrast.	Instructor completes attendance without assistance on laptop/tablet.
UAT / pilot	Workflow correctness and policy fit with real users and approved sample data.	Institute validates formulas, statuses, curriculum mapping and official report format.

17.2 Acceptance criteria for MVP

- An authorized administrator can configure a trade, curriculum version, batch, academic session, student roster and instructor mapping.
- An assigned instructor can create a session and submit a complete attendance register; duplicate records are blocked.
- Late attendance changes require the approved process and preserve who changed what, when and why.
- An instructor can record multiple practical attempts for a student without overwriting previous attempts.
- An authorized assessor can evaluate a competency against configured criteria and record an outcome with feedback.
- The student sees only their own attendance, activities, competencies, assessments and remedial actions.
- An instructor, HOD and administrator see different data scopes matching the approved role matrix.
- Attendance, competency and intervention reports match independently verified sample calculations.
- Exports include applied filters and timestamps and cannot bypass the user's permission scope.
- Critical and high security findings are resolved, backup is configured, and a restore test is demonstrated before production launch.

17.3 Sample UAT scenarios

Scenario	Expected result
Instructor marks a scheduled batch session.	Only active students in the selected batch appear; each receives one valid status; save is confirmed.
Student attempts to open another student's progress URL.	Access is denied; no data is disclosed; attempt is logged where appropriate.
Student completes an activity but has not passed assessment criteria.	Activity may show completed, while competency remains In Progress / Needs Practice per configured rule.
Assessor records a failed criterion and assigns follow-up.	Assessment attempt remains in history; competency shows Needs Practice; intervention has owner and due date.
Admin exports a trade summary.	Only authorized trade scope is included; report displays period, filters, generated timestamp and defined denominators.
A student transfers to another batch.	Historical attendance and training remain linked to the previous enrollment; new sessions use the active enrollment.
A curriculum rubric is updated for a new session.	Historical assessments remain linked to the prior rubric version.

18 | Risks, dependencies and mitigations

Risk / dependency	Potential impact	Mitigation / control
Inconsistent or incomplete student data	Duplicate accounts, incorrect rosters and unreliable reports.	Data profiling, unique identifier rules, import validation, reconciliation and named data owner.
Unclear attendance policy	Disputed percentages and inconsistent attendance status.	Approve status mapping, denominator, exclusions, correction window and examples before coding.
Curriculum variation between trades	Rigid data model or wrong competency mapping.	Use versioned, configurable masters; validate with HODs from representative trades.
Inconsistent trainer assessment	Competency results not comparable across instructors.	Define rubrics, assessor guidance, moderation and calibration sessions.
Low instructor adoption	Parallel registers continue and data becomes incomplete.	Co-design workflows, minimize clicks, train users, pilot with champions and retire duplicate processes only after sign-off.
Poor connectivity or device availability	Delayed capture or incomplete use in workshops.	Responsive design, autosave where safe, printable contingency register and evaluate offline capture as a later phase.
Over-permissive data access	Student privacy breach or inappropriate changes.	Least privilege, scope-based authorization, access review, audit and export restrictions.
Evidence upload misuse	Malware, excessive storage or exposure of personal files.	Private storage, type/size limits, scanning, retention rules and authorized download.
Dashboard metrics misinterpreted	Students or staff judged using incomplete/incorrect indicators.	Publish definitions, denominators, freshness and human-review guidance.
Integration or hosting dependency	Schedule delays or unstable production operation.	Confirm hosting and external system constraints in discovery; isolate integrations behind adapters.
Scope expansion	MVP delivery slips due to adding unrelated features.	Use phased roadmap, change-control log and Must/Should/Could prioritization.
Backup exists but cannot be restored	Prolonged outage or permanent data loss.	Perform and document restore drills, not only backup-job monitoring.

18.1 Critical dependencies to resolve early

- Authoritative student roster, admission identifier and batch assignment source.
- Current trade list, curriculum versions, modules, practical activity and competency criteria.
- Official attendance categories, percentage rules, period definitions and approval route for corrections.
- Assessment terminology, rubric ownership, mandatory criteria and finalization authority.
- Approved user role matrix, data visibility and monitoring report scope.
- Hosting environment, data residency/security requirements, connectivity and device availability.
- Operational ownership, support contact, retention schedule and backup/recovery target.

19 | Pilot, rollout and change management

19.1 Pilot design

Pilot the MVP with a representative but manageable cohort: at least one trade, one or more batches, a small group of instructors and a defined student sample. Where feasible, include a trade with substantial practical work so the competency workflow is validated, not only attendance.

Pilot stage	Activities	Evidence to collect
Prepare	Confirm cohort, clean data, configure trade/curriculum, create accounts, publish role matrix and train instructors.	Approved roster; curriculum mapping; training attendance; access test.
Run	Capture attendance and practical activity; assess selected competencies; use student portal and reports.	Completed sessions; practical logs; assessment examples; student feedback.
Review	Compare records to approved source/parallel sample; verify calculations; review defects and adoption.	Reconciliation report; KPI baseline; defect log; user feedback.
Decide	Fix critical issues, approve process changes, confirm support and sign off broader deployment.	Pilot acceptance; go/no-go decision; prioritized backlog; rollout plan.

19.2 Training and adoption plan

- Administrator training: master data, user provisioning, imports, reports, correction approvals and audit review.
- Instructor training: daily register, practical log, assessment rubric, feedback and intervention workflow.
- Student orientation: sign-in, progress interpretation, privacy, pending tasks and correction request route.
- HOD training: dashboard interpretation, trade-level monitoring, exception approval and action follow-up.
- Provide quick-reference guides, short task videos or demonstrations, and a named support contact during the pilot.
- Gather feedback after initial use and after a complete attendance/practical/assessment cycle.

19.3 Go-live readiness checklist

- Policy and workflow decisions approved; roles and data scope signed off.
- Production environment, HTTPS, database access and private file storage configured.
- Initial master data imported and reconciled; student accounts activated securely.
- Critical functional, security, performance and backup tests passed.
- Users trained; support and escalation route communicated.
- Reports validated against a trusted sample; monitoring dashboards show data freshness.
- Rollback / contingency process and first-week issue triage owner identified.

20 | Open decisions and glossary

20.1 Decisions required before development baseline

Decision	Why it matters	Suggested owner
What is the authoritative academic session and batch structure?	Determines master records, enrollment, reporting periods and student cohort scope.	Institute administrator / academic office
Which attendance statuses are allowed and how are they counted?	Determines attendance percentage and official register outputs.	Institute leadership / attendance owner
What is the correction lock window and approval authority?	Controls integrity and disputes over past attendance.	Institute leadership / HOD
Which curriculum version and competencies apply to each trade/batch?	Determines the training and competency catalogue.	Trade HOD / academic owner
What assessment scale and pass rule are used?	Determines whether competency status can be calculated consistently.	Trade HOD / assessor lead
Can students submit evidence, or do trainers record it only?	Affects workflow, storage, privacy and support.	Academic owner / privacy lead
What personal data is necessary and how long is it retained?	Affects privacy, database design and archival.	Institute leadership / data owner
Which reports are mandatory and who can receive them?	Affects dashboard scope, exports and access control.	Monitoring/report owner
Single institute deployment or multi-institute platform?	Affects tenant isolation, user administration and infrastructure.	Project sponsor / IT owner
What hosting and connectivity constraints apply?	Affects deployment, offline needs, backup and operational support.	IT / hosting owner

20.2 Glossary

Term	Meaning in this blueprint
ITI	Industrial Training Institute.
Trade	A vocational training discipline or programme configured in the institute.
Batch	A cohort of students grouped for training in a trade and academic period.
Curriculum version	A dated or approved version of the modules, competencies and criteria used for a cohort.
Practical activity	A defined task or exercise that a student performs during practical training.
Competency	A defined skill outcome that a student must demonstrate against stated criteria.
Assessment rubric	A structured set of criteria and evaluation levels used by an assessor.
Evidence	An observation, checklist, result or authorized attachment supporting a training/assessment record.
Intervention	A documented remedial support action with an owner, due date and review outcome.
RBAC	Role-Based Access Control: permissions assigned through defined user roles.
MVP	Minimum Viable Product: the smallest usable release that meets the agreed core scope.

Term	Meaning in this blueprint
UAT	User Acceptance Testing: validation by representative users against approved requirements.

Conclusion and recommended next step

ITI SkillTrack should be delivered as a configurable, role-based training record system with a single traceable workflow connecting attendance, practical activity, competency assessment, student visibility and instructor-led remediation. The highest implementation risk is not the choice of programming language; it is ambiguity in attendance policy, curriculum mapping, assessment rules, role boundaries and source data. Those decisions should be validated with institute stakeholders before the development baseline is frozen.

Recommended immediate actions

1. Nominate one product owner and one academic/curriculum owner for each participating trade.
2. Collect a sample student roster, current attendance register, practical activity list, competency/assessment sheet and institutional report format.
3. Approve attendance statuses, percentage denominator, correction workflow, competency scale and assessment finalization rule.
4. Validate the proposed role/permission matrix and decide whether the first release is single-institute or multi-institute.
5. Convert the Must requirements into screen-level acceptance criteria and finalize the MVP backlog.

PROJECT DEFINITION OUTCOME

Once these decisions are signed off, this blueprint can be converted into a detailed SRS, UI/UX prototype, database schema, API specification, sprint backlog and implementation plan.

Appendix A | Worked example: one student journey

The following example is fictional and is included to explain the intended record flow. Replace the sample trade, tasks, criteria and policy values with approved institute data.

A.1 Example progress snapshot

Field	Illustrative value
Learner	Student A (fictional)
Trade / batch	Electrician / Batch ELC-2026-A (illustrative)
Attendance period	Selected review period
Eligible scheduled sessions	20
Attended sessions	18
Illustrative attendance	18 / 20 x 100 = 90% (assuming the institute-approved rule counts these 18 sessions as attended)
Practical activity	Safe wiring circuit assembly - 2 attempts recorded
Competency summary	2 competencies Competent; 1 competency Needs Practice
Open support action	Additional practice on testing and inspection; assigned trainer; review date configured by instructor

A.2 How the system updates the journey

1. Instructor opens the assigned batch session and records the learner as present. The session is submitted and the attendance record is stored with the instructor and timestamp.
2. The learner performs the safe wiring circuit assembly. The trainer creates practical attempt 1, records observations and feedback, and marks a required criterion for rework.
3. The learner practices again. The trainer records attempt 2 as a separate record; attempt 1 remains available in history.
4. The assessor evaluates mapped competencies against the approved rubric. Two competencies meet all mandatory criteria; one remains Needs Practice because a required testing/inspection criterion is not yet met.
5. The instructor assigns a focused practice action, an owner and a review date. The learner sees the action on the student dashboard; the instructor sees it in the follow-up queue.
6. After reassessment, the assessor records a new attempt. If all mandatory criteria are met, the competency status changes to Competent under the configured rule. The prior assessment remains auditable.

A.3 Example competency decision

Criterion	Attempt 1	Attempt 2	Outcome logic
Prepare work area and follow safety instructions	Met	Met	Mandatory criterion satisfied.
Select tools and materials correctly	Met	Met	Criterion satisfied.
Complete wiring connections to specification	Needs rework	Met	Criterion satisfied on second attempt.
Test and inspect the completed circuit	Not yet demonstrated	Not yet demonstrated	Competency remains Needs Practice until the mandatory criterion is met.
Overall competency status	Needs Practice	Needs Practice	Do not mark Competent while a mandatory criterion is outstanding.

ILLUSTRATIVE ONLY

The activity and criteria above are examples, not an official Electrician trade curriculum or prescribed assessment rubric. A real institute must load approved curriculum content and assessment criteria.

Appendix B | Recommended data-capture templates

Template	Minimum fields to capture
Attendance register	Institute; academic session; trade; batch; date; period/session; student identifier; status; marked by; marked time; submission/lock time; correction reason and approver where applicable.
Practical activity log	Student; trade/batch; module; activity code/title; attempt number; activity date; trainer; status; observation; feedback; linked competency; optional evidence reference.
Competency assessment	Student; competency code/version; criteria; assessment method; assessor; attempt number; criterion-level result; overall outcome; score/grade if applicable; feedback; evidence; finalization time.
Remedial action	Student; trigger/reason; supporting record; skill gap; practice task; assigned trainer; created date; due date; status; review note; reassessment reference; closure outcome.

Template	Minimum fields to capture
Report request / export	Report type; date range; institute/trade/batch scope; selected filters; requested by; generated time; output format; data definition/version; export audit reference.

Appendix C | Example validation rules for forms

- Attendance: batch and session must be valid; student must be actively enrolled for the session date; only one record per student/session; status must be from the approved list.
- Practical log: activity must belong to an active curriculum/module; attempt number is unique for the student/activity sequence; trainer must be authorized for the batch or activity.
- Assessment: assessor must have permission; rubric version must be active for the relevant cohort; each mandatory criterion must have an outcome before finalization.
- Intervention: action, owner and due date are required; closure requires a review note; overdue status is derived from due date and closure state.
- Export: all filters and record scopes are server-validated; downloadable files expire or are access-controlled according to institute policy.

C.1 Suggested bulk-import worksheets

Import dataset	Required mapping fields	Validation / reconciliation
Students	Institute code; admission/student ID; name; status; optional approved contact fields.	Unique ID within institute; no blank mandatory values; duplicate and inactive checks.
Trades and batches	Trade code/name; curriculum version; batch code; academic session; start/end dates.	Trade and session must exist; date ranges valid; batch/trade relationship consistent.
Enrollments	Student ID; batch code; effective-from date; effective-to date if applicable; status.	Student and batch must exist; date ranges must not overlap unexpectedly; transfers retain history.
Staff assignments	Staff account ID; role; trade/batch/module scope; effective dates.	Role must be approved; assignment scope must be valid; privileged access reviewed.
Curriculum masters	Trade; curriculum version; module code; competency code; criteria; practical activity code.	Codes unique within version; mappings complete; source owner and approval/effective date recorded.

C.2 Migration and reconciliation sequence

- Obtain source files from the named data owner and preserve an immutable original copy for reconciliation.
- Profile data for missing identifiers, duplicates, invalid dates, inconsistent trade/batch labels and unsupported status values.
- Map source columns to approved target fields; document every transformation and unresolved exception.
- Load a test batch into a non-production environment, compare row counts and sample records, and obtain owner sign-off.
- Run the approved production import, retain import logs and exception reports, and reconcile totals before enabling live transactions.